

MODULE 17 · DOMAIN 3 - CLOUD TECHNOLOGY AND SERVICES

Integration, Analytics and AI

Connecting components, querying data, and the AI service you are expected to name.

DOMAIN 3 of 4 EXAM WEIGHT 34% MODULE 17 of 26

Integration is about **decoupling**. Analytics is about **where the data already lives**. AI is about **naming the task**.

SNS, SQS and EventBridge

SERVICE	HOW IT WORKS	USE IT WHEN
Amazon SNS	Push, one to many	Many subscribers all need the same message
Amazon SQS	Pull, one consumer per message	Work must be queued and processed once
Amazon EventBridge	Route by rule	Events must reach different targets by content

ON THE EXAM

Broadcast means SNS. Buffer work means SQS. Route by what the event contains means EventBridge.

SNS or SQS?

AMAZON SNS

Publish and subscribe

- A message goes to every subscriber
- Pushed immediately to endpoints
- Email, SMS, Lambda, HTTP, SQS queues
- Nothing is stored waiting for a consumer

AMAZON SQS

A queue of work

- Each message is processed by one consumer
- Consumers pull when they are ready
- Messages persist until processed or expire
- Absorbs bursts so consumers are not overwhelmed

THE TIEBREAKER

Notify several places at once means SNS. Hand work to a worker that will process it later means SQS.

Amazon EventBridge

- ◆ **An event bus that routes events to targets** based on rules you write about the event's content.
- ◆ AWS services publish events to it, including EC2 state changes, so it detects things happening in your account.
- ◆ **It can run on a schedule**, which makes it the standard way to trigger a Lambda function periodically.
- ◆ Targets include Lambda, SNS, SQS, Step Functions and many others.
- ◆ It also receives events from software-as-a-service applications outside AWS.

ON THE EXAM

Notify me when an EC2 instance starts or stops is EventBridge detecting the event, with SNS delivering the message.

AWS Step Functions

- ◆ **Coordinates multiple services into a defined workflow** with steps, branches, retries and error handling.
- ◆ The workflow is visual, so you can see which step failed and where a run stopped.
- ◆ **The signal is a multi-step process** with order, conditions or things that must happen in sequence.
- ◆ It commonly orchestrates several Lambda functions that would otherwise have to call each other.
- ◆ A single trigger firing one function is not orchestration. That is EventBridge or a direct trigger.

ON THE EXAM

Several steps that must happen in order, with retries and error handling, is Step Functions. One step is not.

Analytics questions are usually decided by **where the data already is**, not by what you want to do with it.

The analytics line-up

SERVICE	USE IT WHEN
Amazon Athena	Query data already in S3 using SQL, without loading it anywhere
Amazon Redshift	Sustained heavy analytics over a warehouse you load data into
Amazon Kinesis	Data arriving continuously that must be processed in real time
AWS Glue	Extract, transform and load, plus a catalogue of what your data is
Amazon EMR	Big data frameworks such as Apache Spark and Hadoop
Amazon OpenSearch Service	Searching and analysing logs and operational data
Amazon Quick Sight	Dashboards and business intelligence visualisations

ON THE EXAM

Athena queries S3 in place. Redshift is a warehouse you load. That distinction decides most analytics questions.

Athena or Redshift?

AMAZON ATHENA

Leave the data where it is

- Serverless SQL directly against Amazon S3
- Nothing to provision, nothing to load
- Pay per query, by data scanned
- Best for occasional or ad hoc questions

AMAZON REDSHIFT

Load it into a warehouse

- Data is loaded and stored in the warehouse
- Optimised columnar storage for repeated queries
- Pay for the warehouse, running
- Best for constant, heavy analytical workloads

THE TIEBREAKER

Occasional queries on data already in S3 means Athena. Constant analytics by a whole team means Redshift.

Kinesis and streaming

- ◆ **Amazon Kinesis handles data arriving continuously**, such as clickstreams, telemetry and application logs.
- ◆ The giveaway phrases are real time, streaming, and as the data arrives.
- ◆ **Data Streams** gives you the stream to process yourself. **Data Firehose** delivers it to a destination for you.
- ◆ Firehose commonly lands data in Amazon S3, Redshift or OpenSearch with no code.
- ◆ Batch processing at the end of the day is not streaming. That is Glue, EMR or Batch.

ON THE EXAM

Real time and continuous mean Kinesis. Nightly or scheduled means something else entirely.

A note on Amazon Quick Sight

- ◆ The exam guide writes it **Amazon Quick Sight**, and describes business intelligence dashboards.
- ◆ The product was renamed **Amazon Quick Suite in October 2025**, then shortened to Amazon Quick.
- ◆ The capability is unchanged. Dashboards and visualisations over your data.
- ◆ Answer with QuickSight on the exam. Expect to see Quick in the console.
- ◆ Module 22 covers this and the other renames together.

ON THE EXAM

If a question offers QuickSight for a dashboard requirement, that is still the intended answer.

The AI services are a **vocabulary test**. One task, one service, almost no reasoning required.

The AI service line-up

IF THE TASK IS	THE SERVICE IS
Identify objects, faces or text in images and video	Amazon Rekognition
Understand text: sentiment, entities, key phrases	Amazon Comprehend
Turn text into lifelike speech	Amazon Polly
Turn speech into text	Amazon Transcribe
Translate between languages	Amazon Translate
Extract text and data from scanned documents and forms	Amazon Textract
Build a conversational chatbot	Amazon Lex
Build, train and deploy your own models	Amazon SageMaker AI

ON THE EXAM

Eight rows and the exam quotes them almost word for word. This is the highest return per minute in the module.

Textract or Comprehend?

AMAZON Textract

Gets the text out

- Reads scanned documents, forms and tables
- Preserves structure such as form fields
- Input is an image or PDF
- Output is text and structured data

AMAZON Comprehend

Works out what it means

- Analyses text you already have
- Sentiment, entities, key phrases, language
- Input is text
- Output is meaning about that text

THE TIEBREAKER

A scanned document means Textract first. Understanding the resulting text means Comprehend. They are often used in sequence.

SageMaker AI, Bedrock and Amazon Q

- ◆ **Amazon SageMaker AI** is for building, training and deploying your own machine learning models.
- ◆ The signal is a custom model, or a problem the pre-built services do not cover.
- ◆ **Amazon Bedrock** provides access to foundation models for generative AI applications.
- ◆ **Amazon Q** is a generative AI assistant that answers questions about your data and systems.
- ◆ Amazon Q Business closed to new customers on 31 July 2026, folding into Amazon Quick.

ON THE EXAM

The exam guide now writes Amazon SageMaker AI, not Amazon SageMaker. The service is the same.

Business applications

- ◆ **Amazon Connect** is a cloud contact centre for voice and chat customer service, priced pay as you go.
- ◆ **Amazon Simple Email Service** sends bulk and transactional email at scale.
- ◆ **Amazon WorkSpaces** provides virtual desktops that users connect to from anywhere.
- ◆ **Amazon AppStream 2.0** streams a single application rather than a whole desktop.
- ◆ Remember Amazon Connect is a contact centre. AWS Direct Connect is a network link. Read the whole name.

ON THE EXAM

A whole desktop means WorkSpaces. One application streamed to users means AppStream 2.0.

Reverse lookup

IF THE SCENARIO SAYS	THE ANSWER IS
Decouple components so a slow worker does not block	Amazon SQS
Notify several systems of the same event	Amazon SNS
Run a Lambda function on a schedule	Amazon EventBridge
Coordinate a multi-step process with error handling	AWS Step Functions
Query existing S3 data with SQL, occasionally	Amazon Athena
Process clickstream data as it arrives	Amazon Kinesis
Analyse customer reviews for sentiment	Amazon Comprehend
Read scanned invoices into structured data	Amazon Textract

ON THE EXAM

Eight phrasings covering the three families. The exam reuses this vocabulary with very little variation.

An online shop's checkout writes an order, then calls an inventory service, a payment service and an email service in turn. When the email service is slow, checkouts time out and customers abandon their baskets.

The company wants the checkout to complete immediately and the downstream work to happen reliably afterwards.

Which service should the company introduce?

- | | |
|---|---|
| ☐ A Amazon SNS | ☐ B Amazon SQS |
| ☐ C AWS Step Functions | ☐ D Amazon EventBridge |

◆ PAUSE AND DECIDE

Which service should the company introduce?

A Amazon SNS

Would push the message onward, but SNS does not hold work. A slow subscriber still fails without retry.

C AWS Step Functions

Orchestrates order and branching. The problem here is coupling, not sequencing.

B Amazon SQS

Checkout writes to a queue and returns at once. Workers process at their own pace, and messages persist until done.

D Amazon EventBridge

Routes events to targets by rule. It does not buffer work when a consumer is slow.

THE SIGNAL

Complete immediately and happen reliably afterwards together describe a queue. Buffering is the whole point of SQS.

A company wants an email sent to its operations team and a message written to a logging system and a Lambda function invoked, all whenever a new file lands in an S3 bucket. Every one of the three must receive every event.

The company does not want the file-processing application to know about any of these consumers.

Which service fits this pattern?

- | | |
|---|--|
| ☐ A Amazon SQS | ☐ B Amazon SNS |
| ☐ C AWS Step Functions | ☐ D Amazon Athena |

◆ PAUSE AND DECIDE

Which service fits this pattern?

A Amazon SQS

Each message is processed by one consumer. Three consumers would need three queues and something to fill them.

C AWS Step Functions

Coordinates a sequence. Nothing here needs ordering; all three happen independently.

B Amazon SNS

One message published to a topic reaches every subscriber. Publishers never know who is listening.

D Amazon Athena

Queries data in S3. Unrelated to notifying anyone about anything.

THE SIGNAL

Every consumer must receive every event is the fan-out pattern, and fan-out is what SNS exists for.

An analyst needs to answer occasional questions about three years of application log files that already sit in Amazon S3 in compressed text format. The questions arise a few times a month and are different each time.

The company does not want to run any infrastructure or load the data anywhere.

Which service should the analyst use?

- | | |
|--|---|
| ☐ A Amazon Redshift | ☐ B Amazon Athena |
| ☐ C Amazon EMR | ☐ D Amazon Quick Sight |

◆ PAUSE AND DECIDE

Which service should the analyst use?

A Amazon Redshift

Requires loading the data into a warehouse that then runs continuously, for questions asked a few times a month.

C Amazon EMR

Provisions a cluster for big data frameworks. That is infrastructure the company said it does not want.

B Amazon Athena

Serverless SQL directly on the S3 data, nothing to load or provision, charged per query.

D Amazon Quick Sight

Visualises results in dashboards. It needs a data source underneath rather than being one.

THE SIGNAL

Data already in S3, occasional ad hoc questions, no infrastructure. Three phrases all pointing at Athena.

An insurer receives thousands of scanned claim forms as PDF files. It needs to pull the values out of named fields on each form into a structured record, then determine whether the free-text description of the incident indicates an urgent case.

It wants managed services rather than building its own models.

Which two services should the insurer use? (Choose two.)

- ☐ A Amazon Textract
- ☐ B Amazon Comprehend
- ☐ C Amazon Rekognition
- ☐ D Amazon SageMaker AI
- ☐ E Amazon Transcribe

◆ PAUSE AND DECIDE

Which two services should the insurer use? (Choose two.)

- A** Amazon Textract
Reads scanned documents and extracts named form fields into structured data. Exactly the first requirement.
- B** Amazon Comprehend
Analyses the extracted free text for meaning and sentiment. Exactly the second requirement.
- C** Amazon Rekognition
Analyses images for objects and faces. It is not built for reading form fields from documents.
- D** Amazon SageMaker AI
You would build and train your own models, which the company explicitly does not want.
- E** Amazon Transcribe
Speech to text. There is no audio anywhere in this scenario.

THE SIGNAL

Two steps in sequence: get the text out, then understand it. Textract then Comprehend is the standard pairing.

A retailer wants to analyse clickstream data from its website as visitors browse, so it can update a live dashboard showing which products are being viewed in the last few minutes. The data arrives continuously and must not be processed in nightly batches.

Which service should the retailer use to ingest this data?

- | | |
|--|---|
| ☐ A AWS Glue | ☐ B Amazon Kinesis |
| ☐ C Amazon Athena | ☐ D AWS Batch |

◆ PAUSE AND DECIDE

Which service should the retailer use to ingest this data?

A AWS Glue

Extract, transform and load, typically in batches. The scenario rules batches out explicitly.

C Amazon Athena

Queries data at rest in S3. It does not ingest a live stream.

B Amazon Kinesis

Built for continuously arriving data processed in real time. Clickstream is its headline use case.

D AWS Batch

Runs batch computing jobs. The name alone conflicts with the stated requirement.

THE SIGNAL

Continuously, as visitors browse, last few minutes, not nightly batches. Four separate signals for streaming.

A company records customer service calls and wants to know, without a person listening to them, which calls contained an angry customer. It has thousands of hours of audio and no machine learning specialists.

It wants to use managed services only.

Which two services should the company use, in order? (Choose two.)

- ☐ A Amazon Transcribe
- ☐ B Amazon Comprehend
- ☐ C Amazon Polly
- ☐ D Amazon Translate
- ☐ E Amazon SageMaker AI

◆ PAUSE AND DECIDE

Which two services should the company use, in order? (Choose two.)

- A

 Amazon Transcribe
Converts the recorded speech into text, which nothing else here can do.
- B

 Amazon Comprehend
Runs sentiment analysis on the transcribed text to find angry callers.
- C

 Amazon Polly
Text to speech, which is the wrong direction. The company already has audio.
- D

 Amazon Translate
Translates between languages. No language barrier is mentioned.
- E

 Amazon SageMaker AI
Building custom models, which is ruled out by having no ML specialists and wanting managed services.

THE SIGNAL

Audio in, meaning out, always needs two steps. Transcribe converts, then Comprehend interprets.

What to carry out of this module

- ◆ **Broadcast is SNS. Queue work is SQS. Route by content is EventBridge.**
- ◆ Several steps with order and error handling is Step Functions. One step is not.
- ◆ **Athena queries S3 in place. Redshift is a warehouse you load.** Kinesis is data still arriving.
- ◆ Textract gets text out of documents. Comprehend works out what text means. Often used in sequence.
- ◆ Polly makes speech from text. Transcribe makes text from speech. Do not reverse them.

ON THE EXAM

The AI table is worth more per minute than anything else in this module. Learn it verbatim.